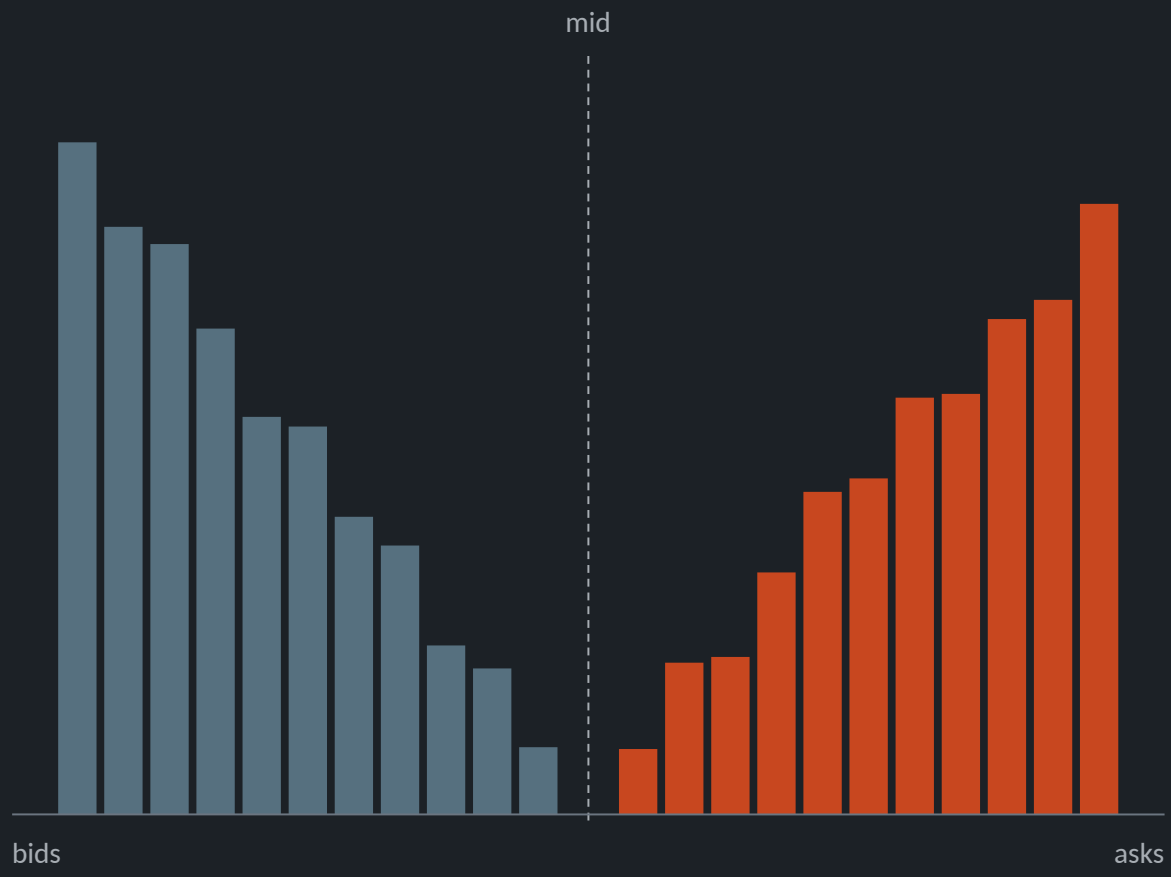
CSE 516 · Probabilistic Graphical Models · Final project (M4)

CausalLOB

What does a sparse temporal graph add to limit-order-book inference?



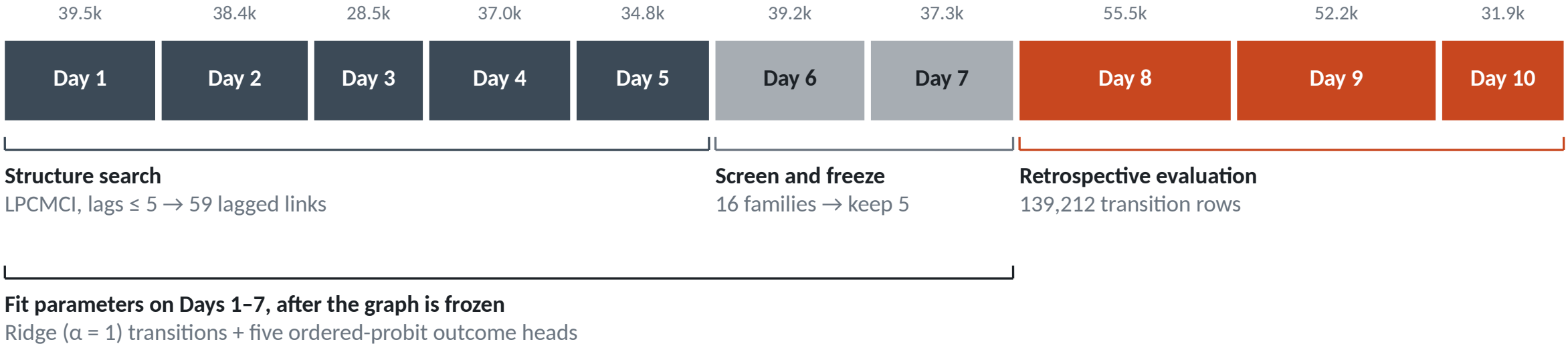
Group G9 · Applied AI

Short answer: useful for narrow inference queries, not as a forecaster

Question	What we found	Verdict
Do nine selected cross-lags add anything beyond each variable's own five lags?	Return RMSE 0.8339 → 0.8247 against AR(5), about 1.1% lower	**Yes, modestly**
Is the sparse graph the best transition model?	Dense VAR(5) wins every pooled transition metric	**No**
Can it reconstruct masked current states?	Gains when return + OFI are hidden; zero or slightly negative elsewhere	**Depends on the mask**
Does it forecast direction better than a neural model?	DeepLOB-style CNN has lower log-loss at all five horizons	**No**
Does it hold up when recent history is missing?	A constant class prior wins all 12 gap tests	**No**

FI-2010: ten trading days, split strictly by time

394,337 order-book snapshots, each with 40 raw price/volume fields and publisher labels at 5 horizons (10, 20, 30, 50, 100 messages). Every day concatenates 5 anonymous securities, giving 50 day–security segments. We respect their boundaries so lags never leak across them, but they are **not** 50 independent experiments.



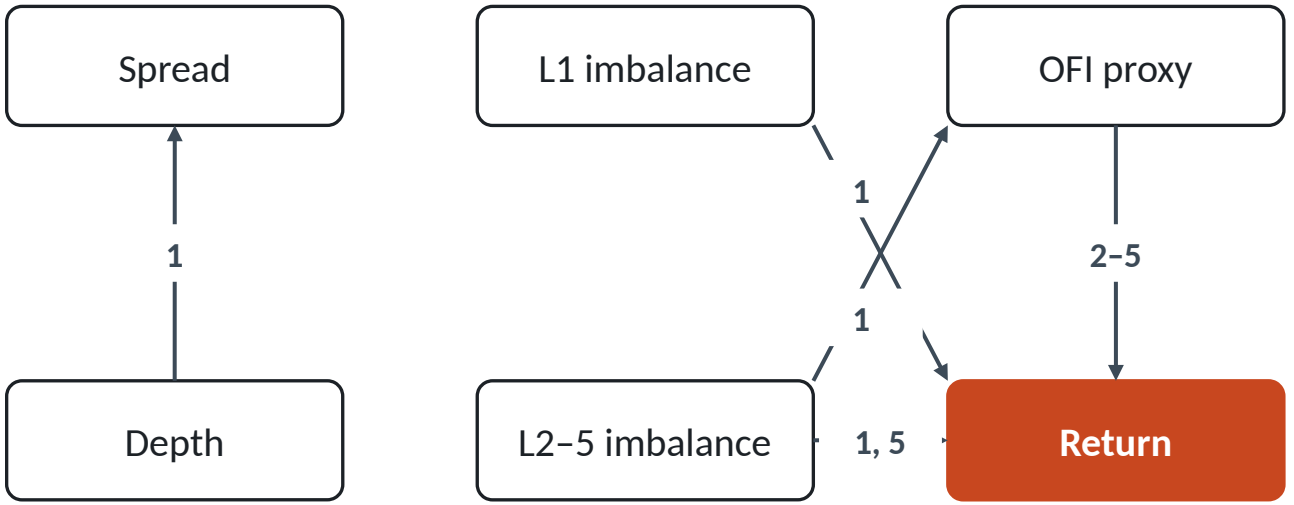
Caveat we keep throughout: Days 8–10 were inspected during development, so every result here is retrospective, not a blind hold-out.

Six state variables, 39 parent slots: a deliberately sparse DBN

Each snapshot t is summarised as

s	log relative spread
d	log aggregate depth, levels 1–5
$q^{(1)}$	level-1 queue imbalance
$q^{(2:5)}$	deeper imbalance, levels 2–5
o	quote-change order-flow (OFI) proxy
r	one-step log mid-price return

OFI here is a quote-change proxy, not executed-order flow.



Arrows are cross-variable lags (labels = lag in snapshots). Every node also keeps lags 1–5 of itself, not drawn.

	Slots	Transition params
AR(5), own history only	30	42
Sparse-39 (ours)	**30 + 9 = 39 (21.7% of 180)**	**51**
Dense VAR(5)	180	192

Inference is exact: Gaussian conditioning, then ordered-probit heads

1 Missing current state

Hide some components m of today's state, observe the rest o plus five lags of history:

$$p(X_{t,m} \mid X_{t,o},\ H_t), \qquad X_t \mid H_t \sim \mathcal{N}(\mu_t,\ Q)$$

$$\mu_{m|o} = \mu_m + Q_{mo} Q_{oo}^{-1} (x_o - \mu_o)$$

$$Q_{m|o} = Q_{mm} - Q_{mo} Q_{oo}^{-1} Q_{om}$$

Closed form, no sampling. Monte Carlo is used only to check the numbers.

2 Future direction

Push the Gaussian state posterior through each horizon's probit score:

$$U_h \mid E \sim \mathcal{N}(\beta_h^{\top}\, \bar{\mu},\ 1 + \beta_h^{\top}\, S\, \beta_h)$$

$$P(\text{Down} \mid E) = \Phi(z_{1h})$$

$$P(\text{Stationary} \mid E) = \Phi(z_{2h}) - \Phi(z_{1h})$$

$$P(\text{Up} \mid E) = 1 - \Phi(z_{2h})$$

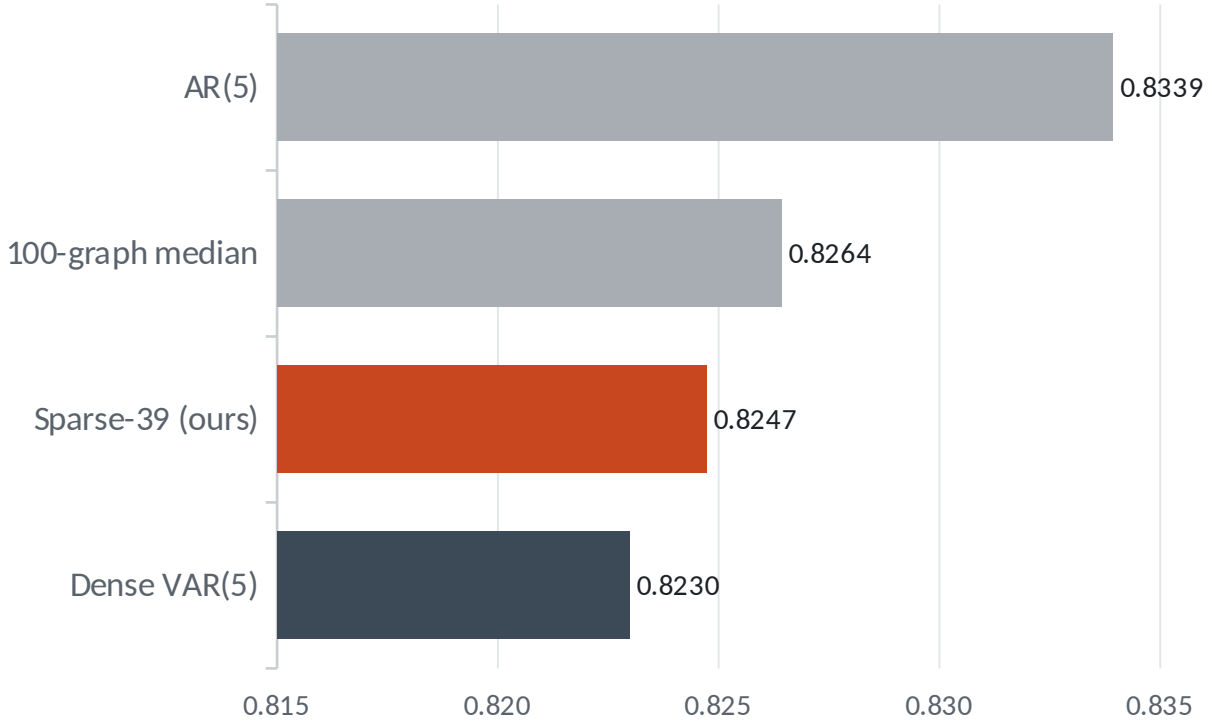
Sums to one by construction, which says nothing about calibration.

$$Q = \text{diag}(\sigma_1^2, \ldots, \sigma_6^2) \ \Rightarrow\ Q_{mo} = 0$$

An observed variable cannot update a missing one at the same step. Only the extra lagged parents can improve a masked prediction.

Cross-lags beat AR(5) by ~1.1%, but dense VAR(5) fits better still

One-step return RMSE, Days 8–10 (lower is better)



Axis starts at 0.815 to make small gaps visible.

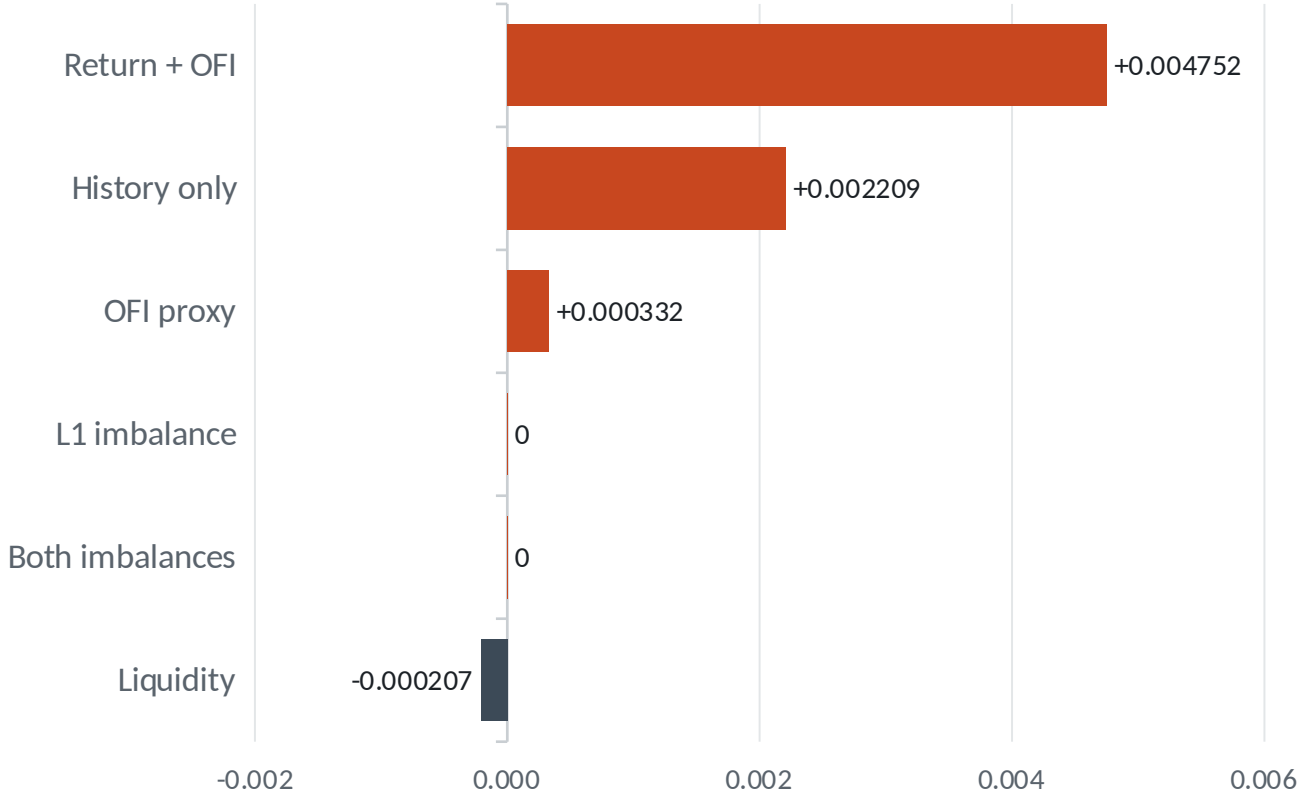
	State RMSE	Joint NLL	Return NLL
AR(5)	.5883	3.7256	1.2650
Sparse-39	.5861	3.7123	1.2520
Dense VAR(5)	**.5852**	**3.7053**	**1.2498**

Sources matter a little. Sparse-39 beats 85–95 of 100 random-source graphs, depending on the metric.

Persistence matters a lot. History vs. no history cuts RMSE by 0.360; the cross-lags add 0.009 on top.

Reconstructing hidden states helps for some masks, not others

RMSE gain over AR(5) by hidden component (positive = Sparse-39 better)



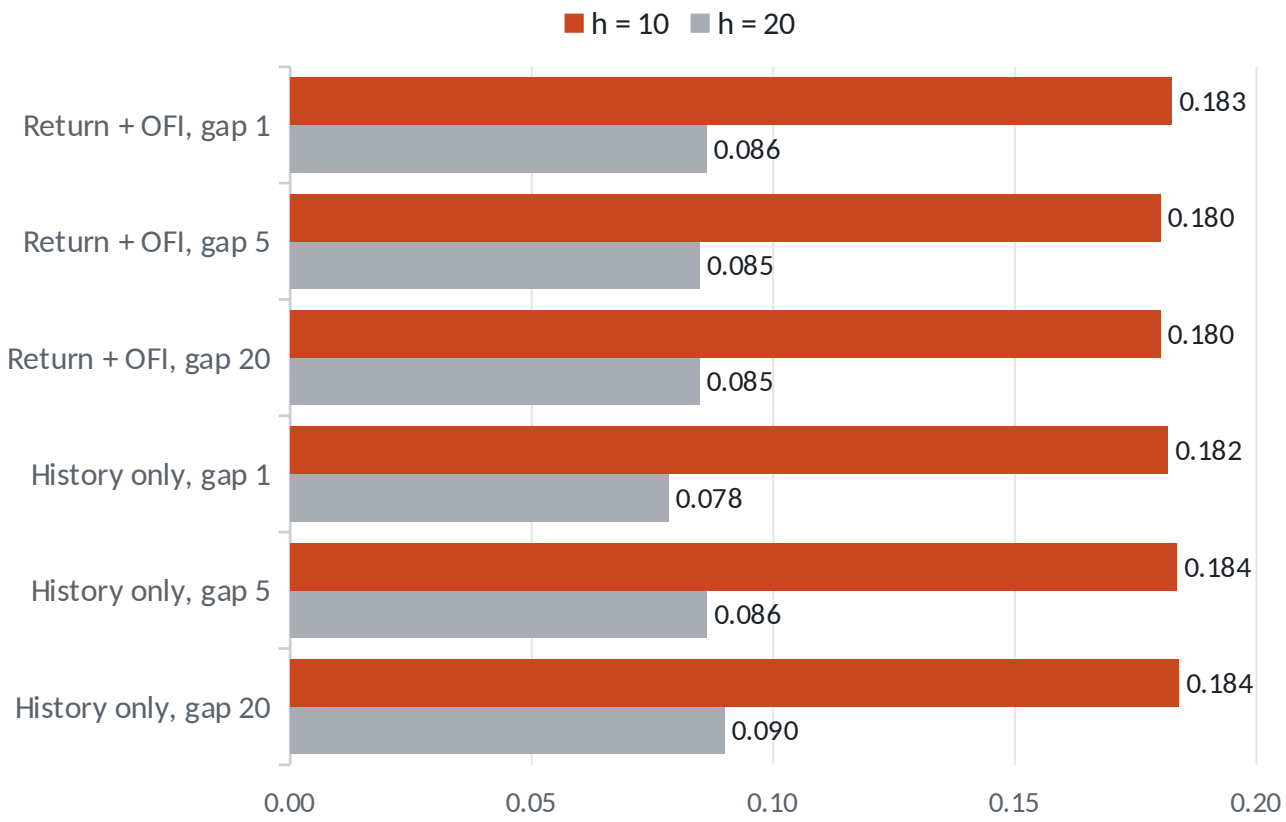
Zero is not a coincidence. The imbalance nodes have no incoming cross-lags, and with diagonal Q the observed variables can't help either. Sparse-39 and AR(5) make identical predictions there.

Better points, worse intervals. For return + OFI, RMSE and NLL improve but 90% coverage error gets worse by 0.006.

Liquidity goes the other way: the depth → spread lag slightly hurts.

After gaps in history, a constant prior beats the DBN in all 12 tests

How much worse the DBN's log-loss is than the prior (DBN − prior)



Setup. 697 spaced endpoints on Days 8–10; 1, 5 or 20 consecutive snapshots withheld; horizons 10 and 20.

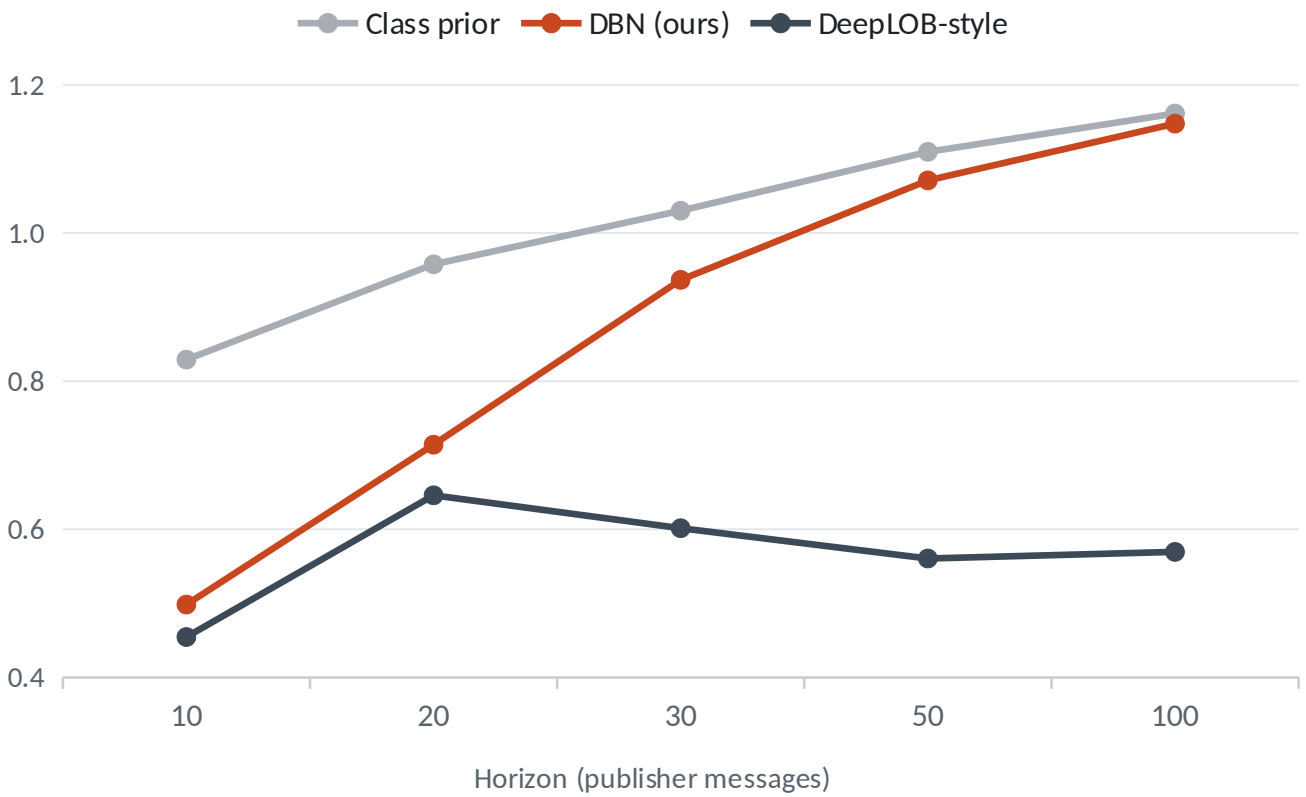
Why it fails. Uncertain states are propagated forward, so the probit head sees inputs further from its training distribution.

What it means. Good one-step reconstruction does not imply reliable forecasts after missing history. A deployed model needs a tested fallback rule.

These are imposed gaps, not real exchange outages.

A DeepLOB-style CNN wins log-loss at every horizon

Three-class log-loss on ~137k matched rows (lower is better)



At h = 10 the metrics disagree

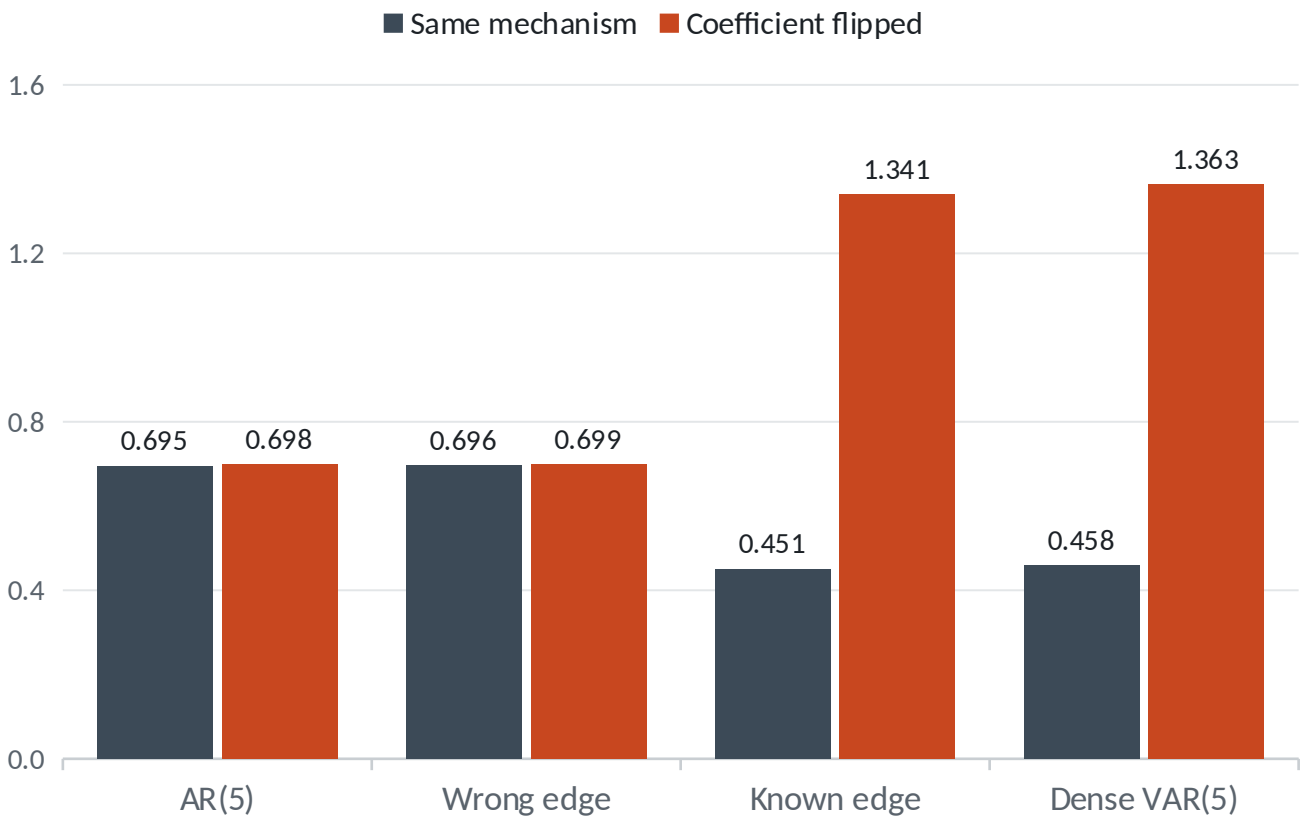
	DBN	DeepLOB
Log-loss	.4984	**.4546**
Brier	**.2368**	.2426
Balanced accuracy	**.7277**	.6763

- Log-loss punishes confident mistakes harder than Brier does.
- DeepLOB is better on 3 of 3 evaluation days at every horizon; with three inspected days this is descriptive, not a significance test.
- Calibration is separate: on full-observation rows the DBN beats the class prior on ECE only at h = 10.

DeepLOB sees raw 100 × 40 windows, so it is a forecasting reference, not an equal-capacity ablation.

A correct edge helps until the mechanism flips, then it hurts most

Synthetic RMSE, mean of 40 replications (lower is better)



Training process

$$x_{5,t} = 0.45\, x_{5,t-1} + 0.80\, x_{0,t-1} + \varepsilon_t$$

Test data keep +0.80 or flip it to −0.80. 800 training and 300 test targets per run; no test-time refitting.

Known edge vs AR(5)	Wins	Mean ΔRMSE [95% CI]
Same mechanism	40 / 40	+0.244 [0.236, 0.253]
Flipped	0 / 40	−0.642 [−0.666, −0.619]

Forty independent runs justify intervals here. They say nothing about market data, and this does not test LPCMCI's graph recovery.

What we can claim, what we can't, and what comes next

Supported

- Nine cross-lags modestly improve return reconstruction over AR(5)
- Exact, inspectable posterior queries for missing states
- Full-evidence Brier below the class prior at all horizons

Not supported

- Causal identification: edges are observational
- A better fit than dense VAR(5)
- Better forecasts than DeepLOB
- Robustness to missing history

Next experiment

- Pre-register comparisons on new days and securities
- Fix a missing-history fallback rule before testing
- Vary training size, noise and regime

CausalLOB is a measured study of interpretable temporal inference, not a new forecasting benchmark.

Limits: ten observational days (three seen during development), anonymous securities, diagonal-Q assumption, imposed rather than real gaps. Reproducibility: 244/244 checkpoint hashes and 320/320 synthetic scores verified by score replay, not full retraining.

Ntakaris et al. 2018 (FI-2010) · Gerhardus & Runge, NeurIPS 2020 (LPCMCI) · Zhang, Zohren & Roberts, IEEE TSP 2019 (DeepLOB) · github.com/aryans7-glitch/CausalLOB-CSE516-2026